

IntelliSecure: A Hybrid AI-Based System for Violence Detection and Predictive Crowd Risk Analysis in Surveillance Videos

Tejas Mestry
Computer Science Engineering
(Data Science)
Vidyavardhini's College Of Engineering
& Technology
(University of Mumbai)
Vasai, India
tejas.234796105@vcet.edu.in

Sukesh Kotian
Computer Science Engineering
(Data Science)
Vidyavardhini's College Of Engineering
& Technology
(University of Mumbai)
Vasai, India
sukesh.234656101@vcet.edu.in

Aditya Kadam
Computer Science Engineering
(Data Science)
Vidyavardhini's College Of Engineering
& Technology
(University of Mumbai)
Vasai, India
aditya.234516102@vcet.edu.in

Sahil Kadge
Computer Science Engineering
(Data Science)
Vidyavardhini's College Of Engineering
& Technology
(University of Mumbai)
Vasai, India
sahil.234526101@vcet.edu.in

Sayalee Susvilkar
Computer Science Engineering
(Data Science)
Vidyavardhini's College Of Engineering
& Technology
(University of Mumbai)
Vasai, India
sayalee.susvilkar@vcet.edu.in

Abstract—With the increasing demand for intelligent public safety systems, traditional surveillance approaches based on passive monitoring are no longer sufficient. This paper presents *IntelliSecure*, a hybrid AI-based surveillance framework designed for automated violence detection and predictive crowd risk analysis. The proposed system integrates a 3D Convolutional Neural Network (3D CNN) for spatio-temporal violence detection with a rule-based crowd analytics module that combines person detection, motion estimation, and density analysis. Unlike conventional methods that treat these tasks independently, *IntelliSecure* provides a unified pipeline capable of detecting multiple risk scenarios from video data. The violence detection module leverages a pretrained R3D-18 architecture to capture motion dynamics across video frames, while the stampede risk module uses YOLOv8 for real-time person detection and optical flow for motion intensity estimation. Experimental evaluation demonstrates that the 3D CNN model outperforms traditional LSTM-based approaches in generalizing to unseen real-world videos. The system is validated on diverse scenarios including sports footage, crowded public environments, and real-life violence clips, showing robust performance and reduced false detections. The proposed approach highlights the effectiveness of combining deep learning with explainable analytical models for real-world surveillance applications.

Keywords— Deep learning, CNN, YOLO, Crowd density estimation, Violence detection, Smart surveillance

I. INTRODUCTION

With rapid urbanization and the rise in large-scale public gatherings, ensuring safety and security in crowded environments has become a significant global challenge. Surveillance systems such as CCTV are widely deployed in public spaces including transportation hubs, stadiums, and commercial areas. However, traditional surveillance systems primarily function as passive monitoring tools, focusing on recording footage rather than actively interpreting scenes or detecting potential threats in real time. This often results in delayed responses during critical situations.

Recent advancements in artificial intelligence (AI) and machine learning (ML) have enabled the development of intelligent video analytics systems that can automatically process and analyze visual data. Computer vision techniques

now support tasks such as object detection, crowd analysis, and behavior monitoring. Despite these advancements, existing systems still face key limitations, including high false alarm rates, limited contextual understanding, and heavy reliance on continuous human supervision. In high-density environments, these limitations become more critical, as risks such as overcrowding, stampedes, and violent incidents require immediate and accurate detection.

The core problem addressed in this work is the lack of an integrated and intelligent surveillance system capable of performing real-time analysis, predicting crowd-related risks, and detecting abnormal or violent activities proactively. Manual monitoring is not only inefficient but also prone to human fatigue, increasing the likelihood of missed events. Additionally, existing automated systems often focus on isolated tasks and fail to provide a comprehensive understanding of dynamic crowd behavior.

To address these challenges, this paper proposes *IntelliSecure*, a context-aware smart surveillance system designed for automated threat detection and crowd risk assessment. The system combines deep learning and computer vision techniques to analyze video streams by leveraging object detection, motion estimation, and behavior-based logic. By integrating both spatial and temporal information, *IntelliSecure* enables early detection of risks such as violence and stampede conditions, reduces dependency on human monitoring, and enhances response efficiency in real-world surveillance scenarios.

II. RELATED WORK

The development of intelligent surveillance systems using Artificial Intelligence (AI) and computer vision has gained significant attention due to the increasing demand for public safety and automated monitoring. Existing research in this domain can be broadly categorized into violence detection, crowd analysis, and risk prediction systems.

A. Violence Detection Approaches

Several studies have focused on detecting violent activities using deep learning models. Mumtaz et al. (2024) proposed a transfer learning-based approach using a deep Convolutional

Neural Network (CNN) with the GoogleNet architecture, achieving an accuracy of up to 99%. However, the system was limited to binary classification and lacked real-time contextual awareness.[1]

Karim et al. (2021) utilized MobileNet and 3D CNN architectures for violence detection, achieving an accuracy of 95.60% on a custom dataset. While the model performed well on low-quality videos, it lacked advanced temporal modeling and real-time decision-making capabilities. [2]

Paul et al. (2025) introduced a lightweight MobileNetV2-based model optimized for edge devices, achieving 91% accuracy. However, the approach relied on frame-level processing, which resulted in the loss of temporal information necessary for understanding complex motion patterns.[4]

Ojha et al. (2024) developed a real-time violence detection system integrated with notification mechanisms using MobileNetV2 and Faster R-CNN. Although the system enabled alert generation, it relied on static threshold-based logic, limiting adaptability and responsiveness.[5]

Additionally, Mahi et al. (2023) introduced the VID dataset, which improved training and generalization capabilities. However, it lacked diverse real-world surveillance scenarios and contextual annotations required for advanced behavior analysis.[6]

B. Crowd Analysis and Stampede Detection

Crowd behavior analysis has been widely explored using both analytical and machine learning approaches. Weng et al. (2024) provided a comprehensive review of crowd risk evaluation methods based on density, velocity, and pressure metrics. While theoretically robust, the study lacked real-time implementation.[7]

Kaur et al. (2023) proposed an IoT-based stampede prediction system using wearable sensors and Support Vector Machine (SVM), achieving an accuracy of approximately 96%. However, the system depended on external devices and lacked visual scene understanding.[8]

Hwang et al. (2024) developed a crowd risk management system using clustering techniques such as DBSCAN to identify high-risk zones. Although effective in simulations, the system lacked validation in real-world scenarios.[9]

Nimbalkar et al. (2025) introduced CrowdSentry, an AI-based system using YOLOv8 and DeepSORT for real-time crowd behavior analysis. While effective, it relied heavily on threshold-based classification and lacked integration with threat detection systems.[10]

Cob-Parro et al. (2021) proposed a stampede detection system using optical flow entropy and ensemble learning methods. Although computationally efficient, it was limited to low-density scenarios and lacked deep contextual understanding.[11]

C. Integrated Surveillance Systems and Limitations

While significant progress has been made in both violence detection and crowd analysis, most existing systems address these problems independently. Ganagavalli et al. (2024) attempted multi-class crime detection using YOLO, CNN, and LSTM, achieving high accuracy. However, the system lacked predictive intelligence and integrated decision-making for proactive surveillance.[3]

Overall, current approaches suffer from several limitations:

- Lack of unified systems combining multiple risk detection tasks
- Limited contextual understanding of dynamic scenes
- Dependence on threshold-based decision mechanisms
- Insufficient validation on real-world surveillance data

These gaps highlight the need for an integrated, context-aware surveillance system capable of combining violence detection with predictive crowd risk analysis, which forms the basis of the proposed IntelliSecure system.

III. CONTRIBUTIONS

The key contributions of this work are as follows:

1. A unified multi-risk surveillance framework that jointly performs violence detection and crowd risk assessment within a single pipeline, addressing the limitation of existing systems that treat these tasks independently.
2. A hybrid intelligence approach that integrates a deep learning-based 3D CNN model with rule-based crowd analytics, enabling a balance between predictive accuracy and interpretability.
3. Implementation and evaluation of a spatio-temporal violence detection model using a pretrained 3D CNN (R3D-18), demonstrating improved generalization on unseen real-world videos compared to traditional CNN-LSTM approaches.
4. A multi-factor crowd risk scoring mechanism that combines person count, motion intensity, and density to enable early detection of potential stampede scenarios.
5. Extensive validation on diverse real-world video scenarios, including sports footage and crowded environments, demonstrating robustness across varying contexts and reducing false detections.

IV. METHODOLOGY

The proposed IntelliSecure system is designed as a unified surveillance framework that processes video streams to detect multiple risk scenarios, including violent activities and crowd-related hazards. The system operates through a multi-stage pipeline consisting of video acquisition, preprocessing, feature extraction, analysis, and decision-making. By combining deep learning with rule-based intelligence, the system achieves both accuracy and interpretability.

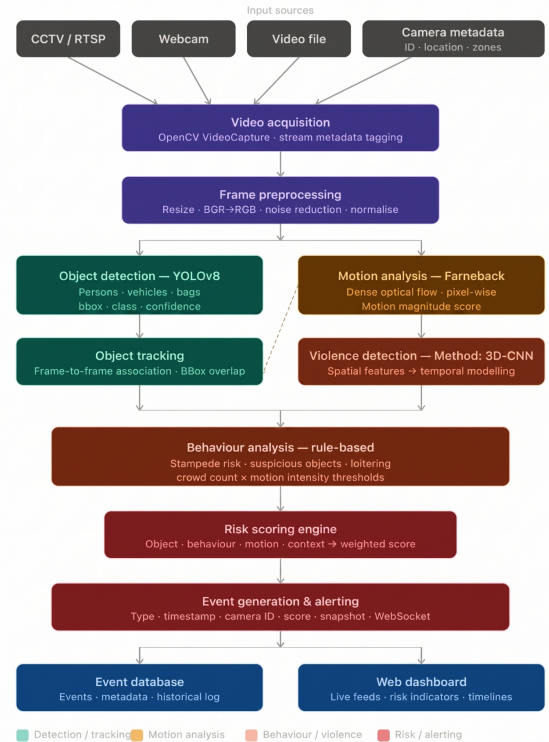


Fig. 1: System Architecture

A. Violence Detection using 3D CNN

To effectively capture spatio-temporal dynamics in video data, a 3D Convolutional Neural Network (3D CNN) is employed for violence detection. Specifically, the R3D-18 architecture pretrained on large-scale video datasets is fine-tuned for binary classification (violence vs non-violence). Unlike traditional approaches such as CNN + LSTM, the 3D CNN performs convolution across both spatial and temporal dimensions simultaneously. This enables the model to directly learn motion patterns such as aggressive interactions, sudden movements, and physical conflicts.

The model processes fixed-length clips of consecutive video frames and outputs a probability score indicating the likelihood of violent activity.

B. Crowd Analysis and Stampede Detection

Crowd behavior is analyzed using a combination of object detection and motion estimation techniques. Person detection is performed using YOLOv8, which provides real-time identification and counting of individuals in each frame.

To capture motion dynamics, optical flow (Farneback method) is used to estimate pixel-level movement between consecutive frames. The magnitude of motion vectors reflects the intensity of crowd movement.

Crowd density is estimated based on the number of detected individuals relative to the frame area. These features are combined to assess crowd conditions and detect potential stampede scenarios.

C. Behaviour Analysis and Risk Scoring

The system incorporates rule-based logic to interpret outputs from detection and motion analysis modules. Multiple behavioral indicators such as crowd density, motion intensity, and object presence are aggregated into a unified risk score.

A weighted scoring mechanism is applied to prioritize critical events while minimizing false positives. High-risk conditions are identified when both crowd density and motion intensity exceed predefined thresholds, indicating potential stampede scenarios. Similarly, high confidence predictions from the violence detection model trigger immediate alerts.

D. Event Generation and System Integration

The final decision layer integrates outputs from all modules to generate event alerts. Each detected event includes contextual information such as event type, timestamp, and confidence score. The system supports both real-time and offline video analysis, enabling flexible deployment across surveillance environments.

This hybrid architecture ensures that IntelliSecure can effectively combine deep learning-based recognition with interpretable rule-based reasoning, making it suitable for practical real-world applications.

V. DATASET AND TRAINING

A. Dataset Description

The violence detection model was trained using the Real Life Violence Situations (RLVS) dataset[12] containing both violent and non-violent scenarios. The dataset includes diverse contexts such as street fights, sports altercations, and normal daily activities to ensure robustness across different environments.

Videos were organized into two primary classes: violence and non-violence. To improve generalization, the dataset

incorporated variations in camera angles, lighting conditions, crowd density, and motion patterns. All data samples were preprocessed to ensure consistency in format and quality before training.

To prevent data leakage and ensure reliable evaluation, the dataset was split at the video level into training and validation sets. This ensures that frames from the same video do not appear in both sets, allowing for a more realistic assessment of model performance on unseen data.

B. Data Preprocessing

Each video was divided into fixed-length clips of consecutive frames. The preprocessing pipeline included:

- Extraction of uniformly sampled frames from each video
- Resizing frames to a fixed resolution of 112×112 pixels
- Normalization of pixel values to match pretrained model requirements
- Formation of clips consisting of 16 consecutive frames

This clip-based representation enables the model to learn temporal patterns effectively.

C. Training Configuration

The violence detection model was implemented using a pretrained R3D-18 (3D CNN) architecture and fine-tuned for binary classification. Transfer learning was employed to leverage learned spatio-temporal features from large-scale video datasets.

The training configuration is summarized below:

- Clip Length: 16 frames
- Image Resolution: 112×112
- Batch Size: 4
- Number of Epochs: 5
- Optimizer: Adam
- Learning Rate: 0.0001
- Loss Function: Cross-Entropy Loss

Training was performed on GPU-enabled hardware to accelerate computation. The model parameters were updated using backpropagation, and gradient-based optimization was used to minimize classification loss.

D. Evaluation Strategy

Model performance was evaluated using a validation dataset consisting of unseen videos. Accuracy was used as the primary evaluation metric. In addition to quantitative evaluation, the model was tested on real-world video scenarios to assess generalization capability.

This included testing on diverse samples such as sports footage, public crowd scenes, and real-life altercations. The evaluation demonstrated that the model effectively distinguishes between violent and non-violent activities across different contexts.

E. Implementation Details

The model was implemented using the PyTorch deep learning framework. Video processing and preprocessing operations were performed using OpenCV. The model was trained on an Apple MacBook Pro equipped with an Apple M3 Pro processor and 18 GB unified memory. PyTorch's

Metal (MPS) backend was used for hardware acceleration where supported.

TABLE I. TRAINING HARDWARE

Hardware	Value
CPU	Apple M3 Pro
RAM	18 GB
Framework	PyTorch
OS	macOS

Overall, the training pipeline was designed to balance computational efficiency with model performance, ensuring suitability for real-time surveillance applications.

VI. RESULTS AND EVALUATION

The proposed 3D CNN-based violence detection model was evaluated on a validation set consisting of 480 unseen videos. The model achieved an overall accuracy of 98.33%, demonstrating strong generalization to unseen data. Performance was further assessed using standard classification metrics. The model achieved a precision of 99.57%, recall of 97.08%, and an F1-score of 98.31%, indicating excellent balance between detecting violent events and minimizing false alarms. Receiver Operating Characteristic (ROC) analysis produced an Area Under the Curve (AUC) of 0.9991, highlighting the model's outstanding discriminative capability. The confusion matrix shown in Figure 5 indicates that only 8 out of 480 validation videos were misclassified, demonstrating robust classification performance across both classes.

TABLE II. MODEL PERFORMANCE COMPARISON

Model	Accuracy	Generalization
ResNet + LSTM	~82%	Poor
3D CNN (R3D-18)	98.33%	Strong

TABLE III. OVERALL 3D CNN MODEL EVALUATION

Metric	Value
Accuracy	98.33%
Precision	99.57%
Recall	97.08%
F1-score	98.31%
AUC	0.9991

TABLE IV. DATASET STATISTICS

Dataset	Videos
Violence	1199
Non-Violence	1200
Training	1919
Validation	480

A. Real-World Scenario Evaluation

To evaluate practical applicability, the system was tested on diverse real-world videos not present in the training dataset. The results demonstrate strong generalization across different contexts:

- MMA fight videos → correctly detected as violent
- Street fight videos → accurately classified as violence
- Hockey fight videos → successfully detected despite different environment
- Non-violent videos (e.g., chess) → correctly classified as safe

Unlike earlier models, the proposed approach effectively captures motion-based patterns rather than relying solely on visual appearance, reducing false negatives.

B. Crowd Risk Analysis Results

The stampede detection module was evaluated using crowd videos with varying densities and motion characteristics.

The system successfully categorized crowd conditions into three levels:

- Low-density + low motion → Safe
- Moderate density → Crowd risk warning
- High density + high motion → High stampede risk

The integration of motion estimation with crowd density significantly improved the detection of abnormal crowd behavior compared to static threshold-based approaches.

C. Qualitative Analysis

Qualitative results demonstrate the effectiveness of the system in real-world scenarios. The system generates bounding boxes for detected individuals, overlays motion-based indicators, and produces risk alerts based on combined analysis.

The results show that the system is capable of identifying both sudden violent events and gradual crowd buildup leading to potential risks. Additionally, the hybrid approach reduces false alarms by incorporating contextual information.

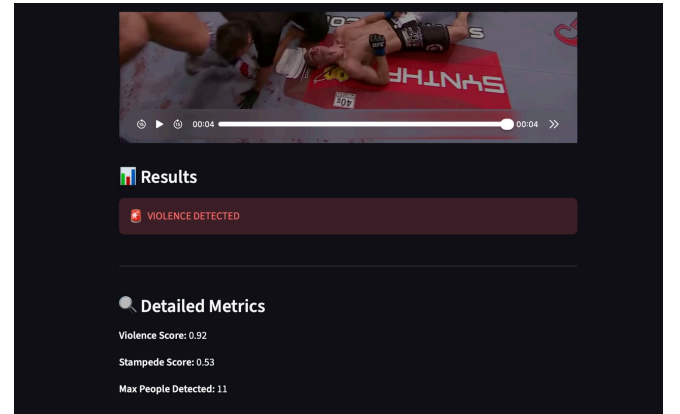


Fig. 2: Violence detection results using the proposed 3D CNN model

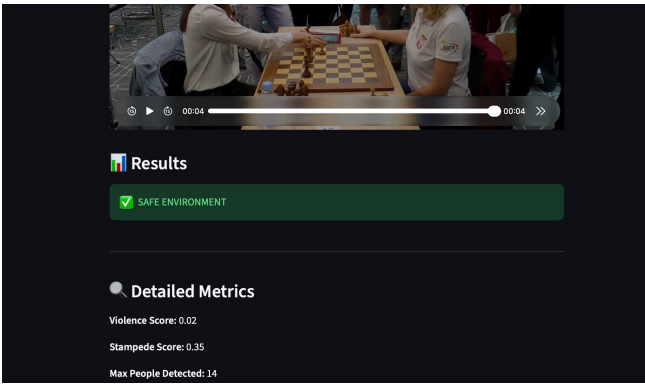


Fig. 3: Safe environment detected in a normal scene

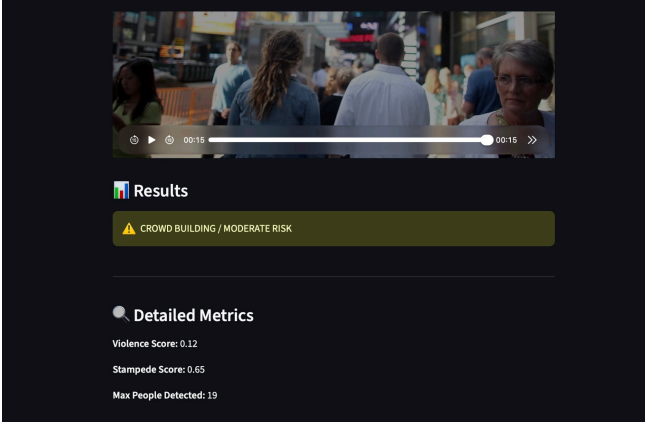


Fig. 4: Crowd risk detection based on density and motion analysis

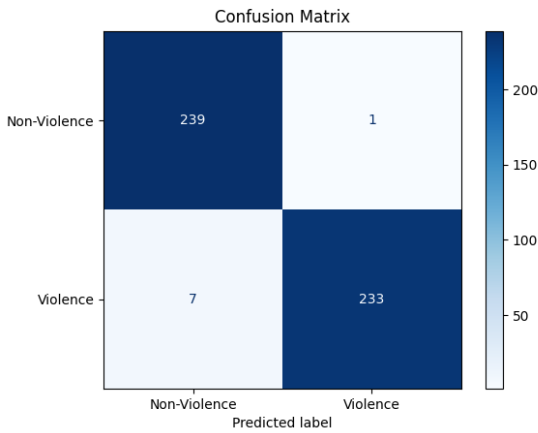


Fig. 5: Confusion Matrix of the Proposed 3D CNN Model

Confusion matrix showing the classification performance of IntelliSecure on the validation dataset. Only eight out of 480 validation videos were misclassified.

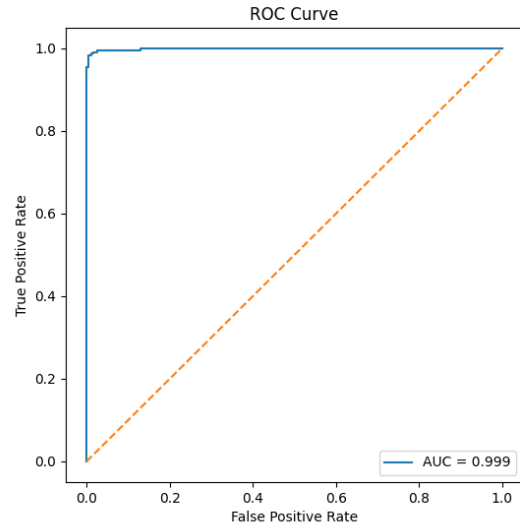


Fig. 6: ROC Curve of the Proposed Model

Receiver Operating Characteristic (ROC) curve for the proposed 3D CNN model, achieving an Area Under the Curve (AUC) of 0.9991.

D. Discussion of Result

The experimental results highlight the importance of spatio-temporal modeling in video understanding tasks. The transition from CNN-LSTM to a 3D CNN architecture resulted in improved performance due to end-to-end temporal feature learning.

Furthermore, combining deep learning with rule-based crowd analytics provides a balanced system that is both accurate and interpretable. This makes IntelliSecure suitable for real-world deployment in surveillance environments.

Although the violence detector achieves high accuracy, the crowd risk module currently uses rule-based reasoning rather than supervised learning. Future work will explore transformer-based video understanding, anomaly detection, and multimodal sensor fusion.

ACKNOWLEDGMENT

The authors express their sincere gratitude to God for guidance, strength, and support throughout the completion of this work. The authors are deeply thankful to Asst. Prof. Sayalee Susvilkar for her invaluable guidance, motivation, and continuous support during the course of this project.

The authors also extend their appreciation to the faculty members of the Department of Computer Science and Engineering (Data Science) for their valuable suggestions and encouragement during the mini project. The support and cooperation provided by the departmental staff are also gratefully acknowledged. Finally, the authors thank their parents for their constant encouragement and support throughout this endeavor.

REFERENCES

- [1] S. Mumtaz, M. Sajjad, and S. W. Baik, "Violence Detection in Surveillance Videos with Deep Network using Transfer Learning," IEEE Access, vol. 8, pp. 55545–55556, 2024.
- [2] A. Karim, M. Hossain, and M. Rahman, "Automatic Violence Detection Technique Using MobileNet," International Journal of Advanced Computer Science and Applications, vol. 12, no. 3, pp. 45–52, 2021.
- [3] P. Ganagavalli and S. Karthikeyan, "YOLO-based Anomaly Activity Detection System for Human Behavior Analysis and Crime Mitigation," Multimedia Tools and Applications, Springer, vol. 83, pp. 12345–12360, 2024.

- [4] S. Paul, A. Das, and R. Roy, "Deep Learning-Based Automated Violence Detection and Identification System for Video Surveillance Applications," *IEEE Transactions on Artificial Intelligence*, vol. 6, no. 1, pp. 112–120, 2025.
- [5] A. Ojha, R. Singh, and P. Sharma, "Enhancing Public Safety: Real-Time Violence Detection and Notification System," *International Journal of Computer Applications*, vol. 185, no. 12, pp. 1–8, 2024.
- [6] A. Mahi, M. Elhoseny, and S. Abdel-Aziz, "VID: A Comprehensive Dataset for Violence Detection in Various Contexts," *Data in Brief*, Elsevier, vol. 49, pp. 109–118, 2023.
- [7] J. Weng, H. Zhang, and Y. Liu, "Review of Analyses on Crowd-Gathering Risk and Its Evaluation Methods," *Safety Science*, Elsevier, vol. 145, pp. 105–115, 2022.
- [8] G. Kaur and R. Singh, "Early Prediction of Stampede in Assemblage," *International Journal of Engineering Research and Technology (IJERT)*, vol. 10, no. 5, pp. 234–240, 2021.
- [9] S. Hwang, J. Lee, and K. Kim, "Crowd-Gathering Risk Management System based on Collision Estimation," *IEEE Access*, vol. 12, pp. 56789–56800, 2024.
- [10] S. Nimbalkar, A. Patil, and R. Kulkarni, "CrowdSentry: An AI-Powered Framework for Real-Time Crowd Behavior Analysis and Stampede Risk Prediction," *IEEE Access*, vol. 13, pp. 10234–10245, 2025.
- [11] A. C. Cob-Parro, C. Losada-Gutiérrez, M. Marrón-Romera, A. Gardel-Vicente, I. Bravo-Muñoz, and M. I. Sarker, "A Proposal on Stampede Detection in Real Environments," *CEUR Workshop Proceedings*, 2021.
- [12] M. M. Soliman, M. H. Kamal, M. A. E. Nashed, Y. M. Mostafa, B. S. Chawky, and D. Khattab, "Violence Recognition from Videos Using Deep Learning Techniques," in *Proc. 2019 9th Int. Conf. Intelligent Computing and Information Systems (ICICIS)*, Cairo, Egypt, 2019, pp. 80–85.